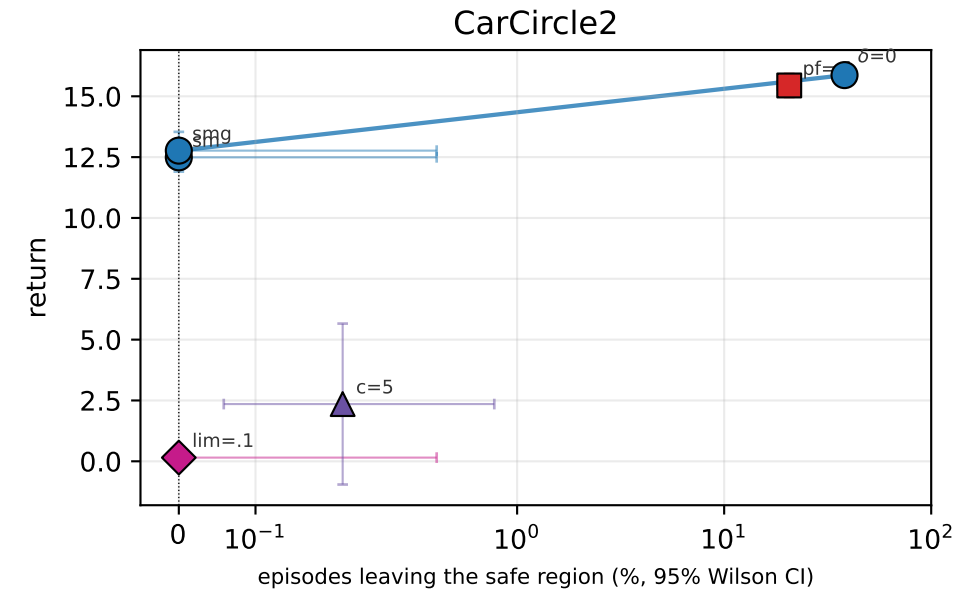


Return vs. violation rate (final checkpoint, 312 episodes/seed; single cost channel; violation = any cost in the episode)



▲ SAC + fixed penalty (penalty-based)   
 ◆ SAC-Lagrangian (penalty-based)   
 ■ FDPI (constraint-based)

● SafeOT-typed (ours)